

Information Architecture

Project Name

CarbonWise

Version: 1.0

Status: Approved

1. Purpose

This document defines how information is organized, structured, navigated, and presented throughout CarbonWise.

The goal is to ensure users can easily:

- Understand their carbon footprint
- Track sustainability progress
- Reduce emissions through personalized actions

while maintaining:

- Accessibility
 - Simplicity
 - Discoverability
 - Scalability
-

2. Information Architecture Goals

Primary Goals

Understand

Help users quickly identify:

- Carbon Score
 - Emission Sources
 - Sustainability Status
-

Track

Help users monitor:

- Trends
 - Goals
 - Historical Data
-

Reduce

Help users discover:

- Personalized Recommendations
 - High Impact Actions
 - Future Opportunities
-

3. Core Navigation Structure

```
graph TD; Home[Home] --- Dashboard[Dashboard]; Home --- CarbonCalculator[Carbon Calculator]; Home --- Recommendations[Recommendations]; Home --- Goals[Goals]; Home --- CarbonTwin[Carbon Twin]; Home --- ActivityHistory[Activity History]; Home --- Profile[Profile]; Home --- Settings[Settings];
```

Home

- Dashboard
- Carbon Calculator
- Recommendations
- Goals
- Carbon Twin
- Activity History
- Profile
- Settings

4. User Journey Architecture

First-Time User

Goal

Understand footprint and receive first recommendation.

Flow

Landing Page

↓

Register

↓

Onboarding

↓

Initial Carbon Assessment

↓

Carbon Score

↓

Recommendations

↓

Goal Creation

↓

Dashboard

Outcome

User understands:

- Current footprint

- Major emission source
 - First action to take
-

Returning User

Goal

Track progress and improve sustainability.

Flow

Login

↓

Dashboard

↓

Review Trends

↓

View Recommendations

↓

Log Activities

↓

Track Goals

↓

Review Carbon Twin

5. Content Hierarchy

Information is prioritized according to challenge requirements.

Level 1 (Highest Priority)

Directly solves problem statement.

Carbon Score

Recommendations

Emission Breakdown

Progress Tracking

Level 2

Supports behavior change.

Goals

Challenges

Activity History

Level 3

Supporting information.

Profile

Settings

Help Center

6. Dashboard Architecture

The dashboard is the most important page.

Dashboard Layout

Dashboard

— Carbon Score

|

— Sustainability Rating

|

— Emission Breakdown

|

— Top Recommendation

|

— Goal Progress

|

— Monthly Trends

|

— Carbon Twin Preview

Carbon Score Card

Displays:

- Current Score
- Monthly Change
- Sustainability Rating

Example

Carbon Score: 68

Rating: Improving

Monthly Change:

+8%

Emission Breakdown

Displays:

Transportation

Food

Energy

Shopping

Visualization

Charts:

- Pie Chart
- Bar Chart

Purpose:

Identify hotspots quickly.

Top Recommendation Widget

Displays:

Highest Impact Recommendation

Includes:

- Estimated Savings
- Difficulty Level
- Quick Action Button

Goal Progress Widget

Displays:

- Goal Completion %
 - Remaining Reduction
 - Next Milestone
-

Carbon Twin Widget

Displays:

What If Preview

Example:

Use Public Transport

↓

Potential Reduction

120 kg CO₂/year

7. Carbon Calculator Architecture

Purpose:

Help users understand emissions.

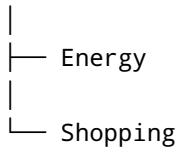
Calculator Structure

Carbon Calculator

├─ Transportation

|

├─ Food



Transportation Section

Collects:

- Vehicle Type
- Distance
- Fuel Type
- Frequency

Food Section

Collects:

- Diet Type
- Weekly Consumption
- Meal Patterns

Energy Section

Collects:

- Electricity Usage
- Appliance Usage
- Heating Usage

Shopping Section

Collects:

- Product Type
- Purchase Frequency
- Spending Category

Output

After submission:

Carbon Score

↓

Category Breakdown

↓

Recommendations

8. Recommendations Architecture

Purpose:

Help users reduce emissions.

Structure

Recommendations

├─ High Impact

├─ Medium Impact

└─ Low Impact

Recommendation Card

Contains:

Title

Example:

Replace Two Car Trips Weekly

Estimated Savings

Example:

120 kg CO₂/year

Difficulty

Options:

Easy

Medium

Hard

Action Button

Example:

Start Goal

Recommendation Prioritization

Order:

Impact

↓

Feasibility

↓

User Preferences

↓

Goal Alignment

9. Goal Architecture

Purpose:

Enable measurable progress.

Structure

Goals

├ Active Goals

├ Completed Goals

└ Milestones

Goal Card

Displays:

- Goal Name
 - Progress %
 - Deadline
 - Estimated Savings
-

Example

Reduce Transportation Emissions

Progress:

40%

Target:

20% Reduction

10. Carbon Twin Architecture

Purpose:

Provide predictive insights.

Structure

Carbon Twin

├─ Current Lifestyle

|

├─ Scenario Builder

|

└─ Forecast Results

Scenario Builder

Users select:

- Transportation Changes
- Food Changes
- Energy Changes

Forecast Results

Displays:

Current Emissions

↓

Projected Emissions

↓

Potential Reduction

Example

Current:

250 kg CO₂

Future:

180 kg CO₂

Savings:

70 kg CO₂

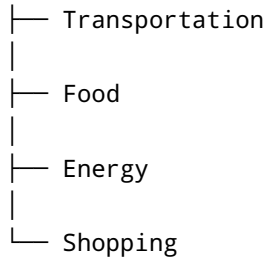
11. Activity History Architecture

Purpose:

Support tracking.

Structure

Activity History



Features

- Filtering
- Search
- Date Range Selection

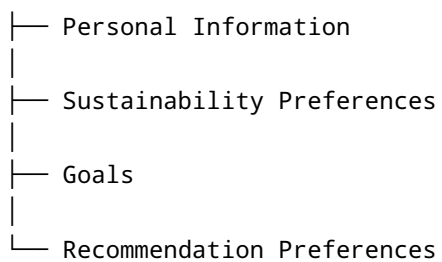
12. Profile Architecture

Purpose:

Enable personalization.

Structure

Profile



Stored Preferences

Examples:

Diet Type

Preferred Transportation

Target Reduction

Notification Settings

13. Mobile Information Architecture

Because most users are mobile-first.

Bottom Navigation

Dashboard

Calculator

Recommendations

Goals

Profile

Mobile Priority Order

1. Carbon Score
 2. Recommendations
 3. Goals
 4. Tracking
-

14. Accessibility Architecture

Supports judging criteria.

WCAG 2.2 AA

Required.

Semantic HTML

Examples:

```
<header>
<nav>
<main>
<section>
<footer>
```

Keyboard Navigation

Supported.

Screen Readers

Supported.

ARIA Labels

Applied to:

- Buttons
 - Forms
 - Charts
-

Color Accessibility

Minimum contrast:

4.5:1

15. Search Architecture

Users can search:

- Recommendations
 - Goals
 - Activities
 - Sustainability Topics
-

16. Personalization Architecture

The interface adapts based on:

- Carbon Score
 - Emission Hotspots
 - Goal Status
 - Recommendation Adoption
-

Example

High Transportation User:

Shows:

Transportation Recommendations First

High Food Impact User:

Shows:

17. Future Expansion

Architecture supports:

- Community Challenges
- AI Carbon Coach
- Smart Device Integration
- Sustainability Marketplace

without major restructuring.

18. Information Architecture Success Criteria

The information architecture is successful when:

- ✓ Users understand their footprint within 60 seconds
- ✓ Users find recommendations within 2 clicks
- ✓ Users can create goals within 1 minute
- ✓ Users can access Carbon Twin simulations easily
- ✓ Accessibility standards are met
- ✓ Navigation remains intuitive
- ✓ The platform directly supports:
 - Understand
 - Track
 - Reduce

objectives from the challenge statement.